

Group Meeting Report

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Bender How To <https://github.com/pulp-platform/bender>

What is Bender: a version management for hardware dependencies and file checklist; encoding dependencies version, arrange codefiles, and generating scripts for different EDA. Deeply used in Pulpissimo.

Typical Start Workflow

1. Save *Bender.yml* in root directory (package checklist)

2. First analysis/update source code by `bender update`

3. Check dependency gplot by `bender packages` & `bender sources -f -t vivado`

4. Generating tool scripts to EDA by:

bender script **vivado** -t vivado > add_files.tcl

bender script **vivado-sim** -t simulation > add_files_sim.tcl

bender script **vsim** -t simulation > compile.tcl

bender script **verilator** -t simulation > verilate.f

5. Lock a reproducible version and submit to repo with *Bender.lock*

Save *Bender.yml* in root directory (package checklist)

```
albert@localhost:~/Works/pulplissimo> bender update -f
warning: Using config at "/home/albert/Works/pulplissimo/.bender.yml" for overrides.
Cloning common_cells (https://github.com/pulp-platform/common_cells.git)
Cloning apb (https://github.com/pulp-platform/apb.git)
Cloning jtag_pulp (https://github.com/pulp-platform/jtag_pulp.git)
Cloning pulp_soc (https://github.com/pulp-platform/pulp_soc.git)
Cloning tbtools (https://github.com/pulp-platform/tbtools.git)
Cloning tech_cells_generic (https://github.com/pulp-platform/tech_cells_generic.git)
Cloning register_interface (https://github.com/pulp-platform/register_interface.git)
Cloning generic_fll (https://github.com/pulp-platform/generic_fll.git)
Cloning apb_fll_if (https://github.com/pulp-platform/apb_fll_if.git)
Cloning common_verification (https://github.com/pulp-platform/common_verification.git)
Cloning adv_dbg_if (https://github.com/pulp-platform/adv_dbg_if.git)
Cloning apb2per (https://github.com/pulp-platform/apb2per.git)
Cloning apb_adv_timer (https://github.com/pulp-platform/apb_adv_timer.git)
Cloning apb_interrupt_cntrl (https://github.com/pulp-platform/apb_interrupt_cntrl.git)
Cloning axi (https://github.com/pulp-platform/axi.git)
Cloning timer_unit (https://github.com/pulp-platform/timer_unit.git)
Cloning cluster_interconnect (https://github.com/pulp-platform/cluster_interconnect.git)
Cloning fpnew (https://github.com/pulp-platform/cvfpnew.git)
Cloning cv32e40p (https://github.com/pulp-platform/cv32e40p.git)
Cloning ibex (https://github.com/pulp-platform/ibex.git)
Cloning scm (https://github.com/pulp-platform/scm.git)
Cloning pulp_io (https://github.com/pulp-platform/pulp-io.git)
Cloning hwpe-mac-engine (https://github.com/pulp-platform/hwpe-mac-engine.git)
Cloning riscv-dbg (https://github.com/pulp-platform/riscv-dbg.git)
Cloning fpu_div_sqrt_mvp (https://github.com/pulp-platform/fpu_div_sqrt_mvp.git)
Cloning udma_uart (https://github.com/pulp-platform/udma_uart.git)
Cloning udma_core (https://github.com/pulp-platform/udma_core.git)
Cloning udma_i2c (https://github.com/pulp-platform/udma_i2c.git)
Cloning udma_sdio (https://github.com/pulp-platform/udma_sdio.git)
Cloning udma_i2s (https://github.com/pulp-platform/udma_i2s.git)
Cloning udma_camera (https://github.com/pulp-platform/udma_camera.git)
Cloning udma_qspi (https://github.com/pulp-platform/udma_qspi.git)
Cloning udma_filter (https://github.com/pulp-platform/udma_filter.git)
Cloning udma_hyper (https://github.com/pulp-platform/udma_hyper.git)
Cloning hwpe-stream (https://github.com/pulp-platform/hwpe-stream.git)
Cloning hwpe-ctrl (https://github.com/pulp-platform/hwpe-ctrl.git)
Checkout tech_cells_generic (https://github.com/pulp-platform/tech_cells_generic.git)
Checkout register_interface (https://github.com/pulp-platform/register_interface.git)
Checkout apb_fll_if (https://github.com/pulp-platform/apb_fll_if.git)
Checkout generic_fll (https://github.com/pulp-platform/generic_fll.git)
Checkout pulp_soc (https://github.com/pulp-platform/pulp_soc.git)
Checkout tbtools (https://github.com/pulp-platform/tbtools.git)
Checkout common_verification (https://github.com/pulp-platform/common_verification.git)
Checkout common_cells (https://github.com/pulp-platform/common_cells.git)
Checkout axi (https://github.com/pulp-platform/axi.git)
Checkout hwpe-stream (https://github.com/pulp-platform/hwpe-stream.git)
Checkout cluster_interconnect (https://github.com/pulp-platform/cluster_interconnect.git)
Checkout scm (https://github.com/pulp-platform/scm.git)
```

If encounter Internet issue, maybe return
`error: Dependency `***` from
`https://github.com/pulp-platform/\*\*\*.git`
cannot satisfy requirement `^0.\*.\*`, pls try:
``
rm -rf .bender/git/db
bender update -f
``

Check dependency by `bender packages`

```
albert@localhost:~/Works/pulpiissimo> bender packages
common_verification
tech_cells_generic
common_cells
apb_axi
fpu_div_sqrt_mvp register_interface udma_core
fpnew gpio hwpe-ctrl hwpe-stream udma_camera udma_filter udma_hyper udma_i2c udma_i2s udma_qspi udma_sdio udma_uart
adv_dbg_if apb2per apb_adv_timer apb_interrupt_ctrl cluster_interconnect cv32e40p hwpe-mac-engine ibex jtag_pulp pulp_io riscv-dbg scm timer_unit
apb_fll_if generic_fll pulp_soc pulpiissimo_optional_vips pulpiissimo_padframe_fpga pulpiissimo_padframe_rtl_sim tbtools
```

Check gplot by `bender sources -f -t vivado`

```
{
  "tech_cells_generic": []
},
"defines": {},
"files": [
  "/home/albert/Works/pulpiissimo/.bender/git/checkouts/udma_qspi-cdef62325988d740/rtl/qspi_pkg.sv",
  "/home/albert/Works/pulpiissimo/.bender/git/checkouts/udma_qspi-cdef62325988d740/rtl/udma_spim_ctrl.sv",
  "/home/albert/Works/pulpiissimo/.bender/git/checkouts/udma_qspi-cdef62325988d740/rtl/udma_spim_reg_if.sv",
  "/home/albert/Works/pulpiissimo/.bender/git/checkouts/udma_qspi-cdef62325988d740/rtl/udma_spim_txrx.sv",
  "/home/albert/Works/pulpiissimo/.bender/git/checkouts/udma_qspi-cdef62325988d740/rtl/udma_spim_top.sv",
  "/home/albert/Works/pulpiissimo/.bender/git/checkouts/udma_qspi-cdef62325988d740/rtl/udma_qspi_wrap.sv"
],
"dependencies": [
  "udma_core",
  "common_cells",
  "tech_cells_generic"
],
"version": "2.0.0"
},
{
  "package": "udma_sdio",
  "independent": false,
  "target": "all(*, all(*))",
  "include_dirs": [],
  "export_incdirs": {
    "udma_sdio": [],
    "udma_core": [],
    "common_cells": [
      "/home/albert/Works/pulpiissimo/.bender/git/checkouts/common_cells-bdd3ba741800de55/include"
    ]
  },
  "tech_cells_generic": []
},
"defines": {},
"files": [
  "/home/albert/Works/pulpiissimo/.bender/git/checkouts/udma_sdio-137248d661f5f240/rtl/sdio_pkg.sv",
  "/home/albert/Works/pulpiissimo/.bender/git/checkouts/udma_sdio-137248d661f5f240/rtl/sdio_crc16.sv",
  "/home/albert/Works/pulpiissimo/.bender/git/checkouts/udma_sdio-137248d661f5f240/rtl/sdio_crc7.sv",
  "/home/albert/Works/pulpiissimo/.bender/git/checkouts/udma_sdio-137248d661f5f240/rtl/udma_sdio_reg_if.sv",
  "/home/albert/Works/pulpiissimo/.bender/git/checkouts/udma_sdio-137248d661f5f240/rtl/sdio_txrx_cmd.sv",
  "/home/albert/Works/pulpiissimo/.bender/git/checkouts/udma_sdio-137248d661f5f240/rtl/sdio_txrx_data.sv",
  "/home/albert/Works/pulpiissimo/.bender/git/checkouts/udma_sdio-137248d661f5f240/rtl/sdio_txrx.sv",
  "/home/albert/Works/pulpiissimo/.bender/git/checkouts/udma_sdio-137248d661f5f240/rtl/udma_sdio_top.sv",
  "/home/albert/Works/pulpiissimo/.bender/git/checkouts/udma_sdio-137248d661f5f240/rtl/udma_sdio_wrap.sv"
],
}
```

Generating tool scripts to EDA by:

```
```\n# bender script <format> -t <target> > <filename>\nbender script vivado -t vivado > add_files.tcl\nbender script vivado-sim -t simulation > add_files_sim.tcl\n\nbender script vsim -t simulation > compile.tcl\n\nbender script verilator -t simulation > verilate.f\n```\n
```

**! Remind to use 'bender update' before generating scripts to lock the dependencies version**

### For Vivado:

```
```\nbender script vivado -t vivado > add_files.tcl\nbender script vivado-sim -t simulation > add_files_sim.tcl\n\n# in Vivado Console, to add/update files\nsource add_files.tcl\n# in Vivado Source Flow, to add/update files\nsource add_files_sim.tcl\n```\n
```

Showcase

Works > pulpissimo > ≡ add_files.tcl

```
723 set_property include_dirs [list \
731 $ROOT/.bender/git/checkouts/cv32e40p-703290f15a6e8974/rtl/include \
732 $ROOT/.bender/git/checkouts/hwpe-ctrl-e547e392627a22fd/rtl \
733 $ROOT/.bender/git/checkouts/hwpe-mac-engine-db1193857e43071d/rtl \
734 $ROOT/.bender/git/checkouts/hwpe-stream-1004699fca3c87e1/rtl \
735 $ROOT/.bender/git/checkouts/ibex-3d636992cd289cea/rtl \
736 $ROOT/.bender/git/checkouts/ibex-3d636992cd289cea/vendor/lowrisc_ip/ip/prim/rtl \
737 $ROOT/.bender/git/checkouts/pulp_io-8557a8d559d27a25/inc \
738 $ROOT/.bender/git/checkouts/pulp_soc-b7e7c62781de8fd8/rtl/include \
739 $ROOT/.bender/git/checkouts/register_interface-d81701e89d22e79e/include \
740 $ROOT/.bender/git/checkouts/udma_core-55b32791e708906d/rtl \
741 $ROOT/.bender/git/checkouts/udma_core-55b32791e708906d/rtl/common \
742 $ROOT/.bender/git/checkouts/udma_hyper-25d0d36a772cf85e/udma-hyperbus/src \
743 $ROOT/.bender/git/checkouts/udma_i2c-a768c09ae26d9daa/rtl \
744 $ROOT/.bender/git/checkouts/udma_qspi-cdef62325988d740/rtl \
745 $ROOT/hw/includes \
746 $ROOT/hw/padframe/pulpissimo_padframe_fpga_autogen/include \
747 $ROOT/hw/padframe/pulpissimo_padframe_rtl_sim_autogen/include \
748 ] [current_fileset -simset]
749
750
751 v set_property verilog_define [list \
752     TARGET_FPGA \
753     TARGET_SYNTHESIS \
754     TARGET_VIVADO \
755     TARGET_XILINX \
756 ] [current_fileset]
757
758 v set_property verilog_define [list \
759     TARGET_FPGA \
760     TARGET_SYNTHESIS \
761     TARGET_VIVADO \
762     TARGET_XILINX \
763 ] [current_fileset -simset]
764
```

Works > pulpissimo > ≡ add_files_sim.tcl

```
830 ] [current_fileset -simset]
831
832 $ROOT/.bender/git/checkouts/cv32e40p-703290f15a6e8974/rtl/include \
833 $ROOT/.bender/git/checkouts/hwpe-ctrl-e547e392627a22fd/rtl \
834 $ROOT/.bender/git/checkouts/hwpe-mac-engine-db1193857e43071d/rtl \
835 $ROOT/.bender/git/checkouts/hwpe-stream-1004699fca3c87e1/rtl \
836 $ROOT/.bender/git/checkouts/ibex-3d636992cd289cea/rtl \
837 $ROOT/.bender/git/checkouts/ibex-3d636992cd289cea/vendor/lowrisc_ip/ip/prim/rtl \
838 $ROOT/.bender/git/checkouts/pulp_io-8557a8d559d27a25/inc \
839 $ROOT/.bender/git/checkouts/pulp_soc-b7e7c62781de8fd8/rtl/include \
840 $ROOT/.bender/git/checkouts/register_interface-d81701e89d22e79e/include \
841 $ROOT/.bender/git/checkouts/udma_core-55b32791e708906d/rtl \
842 $ROOT/.bender/git/checkouts/udma_core-55b32791e708906d/rtl/common \
843 $ROOT/.bender/git/checkouts/udma_hyper-25d0d36a772cf85e/udma-hyperbus/src \
844 $ROOT/.bender/git/checkouts/udma_i2c-a768c09ae26d9daa/rtl \
845 $ROOT/.bender/git/checkouts/udma_qspi-cdef62325988d740/rtl \
846 $ROOT/hw/includes \
847 $ROOT/hw/padframe/pulpissimo_padframe_fpga_autogen/include \
848 $ROOT/hw/padframe/pulpissimo_padframe_rtl_sim_autogen/include \
849 ] [current_fileset -simset]
850
851
852 set_property verilog_define [list \
853     TARGET_FPGA \
854     TARGET_SIMULATION \
855     TARGET_VIVADO \
856     TARGET_XILINX \
857 ] [current_fileset]
858
859
860 set_property verilog_define [list \
861     TARGET_FPGA \
862     TARGET_SIMULATION \
863     TARGET_VIVADO \
864     TARGET_XILINX \
865 ] [current_fileset -simset]
866
```

Developing Process

This is a how to in developing and maintenance: **Only Edit `Bender.yml`!**

1. Developing in separate ip repo `axi/` or `gpio/`

2. Tag and push (must vX.Y.Z):
git tag -a v1.12.0 -m "axi: release 1.12.0"
git push origin v1.12.0

3. Edit top `Bender.yml` and git push in soc repo:
dependencies:

axi:

git: "git@github.com:your-org/axi.git"

version: "=1.12.0" # accurate v1.12.0

or "~1.12" # allow 1.12 or near more

or ">=1.11,<2.0" # allow 1.x newest, allow semver

5. `bender update` and `bender checkout`

Localization dependency

```
dependencies:
  common_cells:      { git: "https://github.com/pulp-platform/common_cells.git",      version: 1.21.0 }
  apb:               { git: "https://github.com/pulp-platform/apb.git",              version: 0.2.4 }
  jtag_pulp:         { git: "https://github.com/pulp-platform/jtag_pulp.git",        version: 0.2.0 }
  pulp_soc:          { git: "https://github.com/pulp-platform/pulp_soc.git",        version: 5.0.1 }
  tbtools:           { git: "https://github.com/pulp-platform/tbtools.git",         version: 0.2.1 }
  tech_cells_generic: { git: "https://github.com/pulp-platform/tech_cells_generic.git", version: 0.2.3 }
  pulpissimo_padframe_rtl_sim: { path: "hw/padframe/pulpissimo_padframe_rtl_sim_autogen" }
  pulpissimo_padframe_fpga: { path: "hw/padframe/pulpissimo_padframe_fpga_autogen" }
  register_interface: { git: "https://github.com/pulp-platform/register_interface.git", version: 0.4.1 }
  generic_FLL:       { git: "https://github.com/pulp-platform/generic_FLL.git",      version: 0.2.0 }
  apb_fll_if:        { git: "https://github.com/pulp-platform/apb_fll_if.git",      version: 0.2.1 }
```

- Change to GitLab in order to private version management and developing

Next Week

1. Continue Localization dependency
2. Environment Configuration
3. Bender